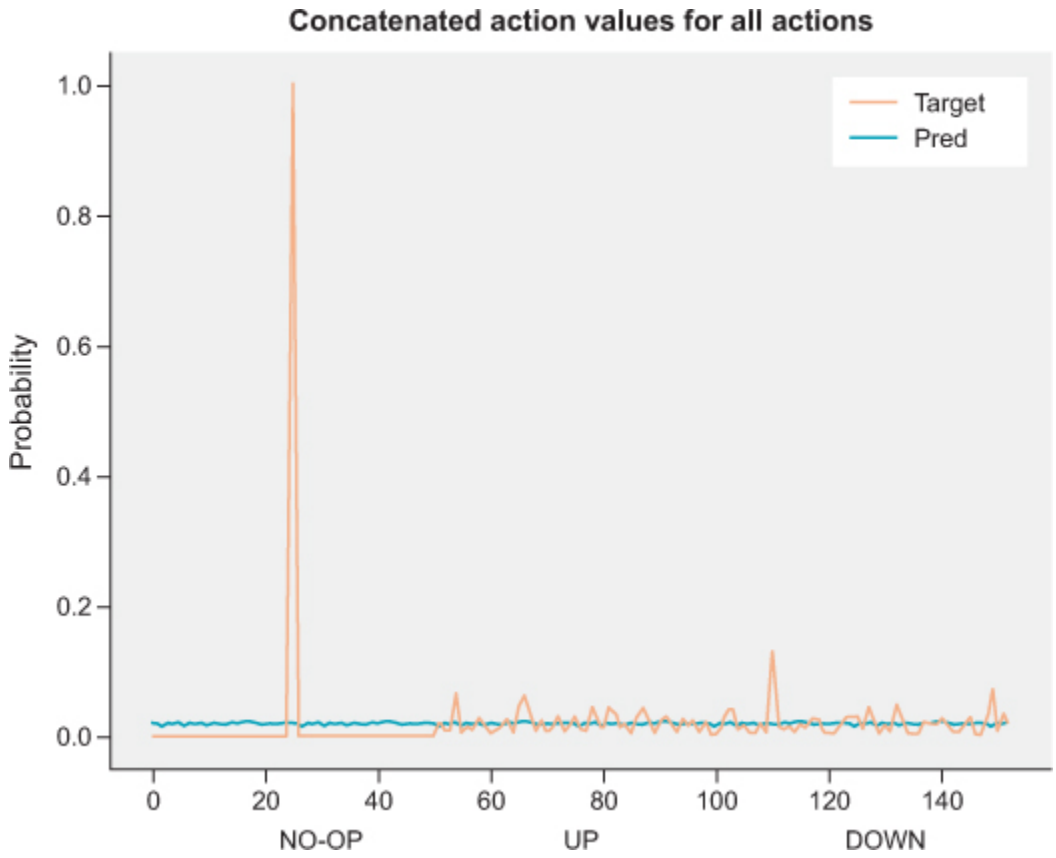


The reason why a target network is so important is very clear with Dist-DQN. Remember, a target network is just a copy of the main model that we update after some lag time. We use the target network's prediction to create the target for learning, but we only use the main model parameters to do gradient descent. This stabilizes the training because without a target network the target distribution will change after each parameter update from gradient descent.

Yet gradient descent is trying to move the parameters toward better matching the target distribution, so there is a circularity (hence instability) that can lead to the target distribution dramatically changing as a result of this dance between the Dist-DQN's predictions and the target distribution. By using a lagged copy of the Dist-DQN prediction (via a lagged copy of the parameters, which is the target network), the target distribution does not change every iteration and is not immediately affected by the continual updates from the main Dist-DQN model. This significantly stabilizes the training. If you reduce the `update_rate` to 1 and try training, you will